

HOMEWORK 5
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

SECTION 2.1 - CONTOUR INTEGRALS

1. Evaluate the following:

- (a) $\int_{\gamma} y dz$, where γ is the union of the line segments joining 0 to i and then i to $i + 2$.
- (b) $\int_{\gamma} \sin 2z dz$, where γ is the line segment joining $i + 1$ to $-i$.
- (c) $\int_{\gamma} z e^{z^2} dz$, where γ is the unit circle.

2. Is it true that $\operatorname{Re}\left(\int_{\gamma} f dz\right) = \int_{\gamma} \operatorname{Re} f dz$?

3. Evaluate $\int_{\gamma} x^2 - y^2 dz$, where γ is the straight line from 0 to i .

4. Let γ be the arc of the circle $|z| = 2$ that lies in the first quadrant. Show that

$$\left| \int_{\gamma} \frac{1}{z^2 + 1} dz \right| \leq \frac{\pi}{3}.$$

SECTION 2.2 - CAUCHY'S THEOREM - A FIRST LOOK

5. For what simple closed curves γ does the equation $\int_{\gamma} \frac{dz}{z^2 + z + 1} = 0$ hold?

6. Let γ_1 be the circle of radius 1 and let γ_2 be the circle of radius 2 (traversed counterclockwise and centered at the origin). Show that

$$\int_{\gamma_1} \frac{dz}{z^3(z^2 + 10)} = \int_{\gamma_2} \frac{dz}{z^3(z^2 + 10)}.$$

7. Evaluate $\int_{\gamma} \frac{2z^2 - 15z + 30}{z^3 - 10z^2 + 32z - 32} dz$ where γ is the circle $|z| = 3$. Hint: Use partial fractions; one root of the denominator is $z = 2$.